

Tech & Teach

Digital Chip Design Program

RTL Design

Almas Alshubrumi

Atheer Al-harbi

Mjd AL-harbi

Maha Omar

Jood Alomair

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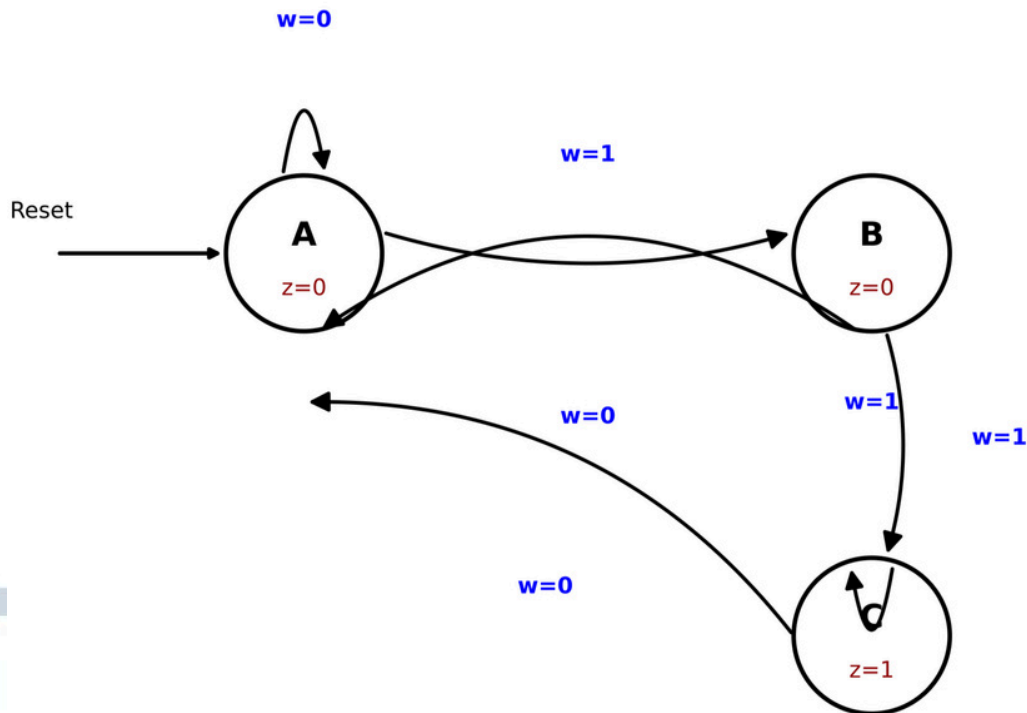
Controller – Part 1

Version 1.0

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Task #12.1

State Diagram - module simple (Moore FSM)



This is a Moore FSM because the output logic block is sensitive only to the current state y , not to the input w . Since z is determined purely by the state and not directly by the input, this matches the definition of a Moore machine, where outputs depend only on the current state.

Task #12.2

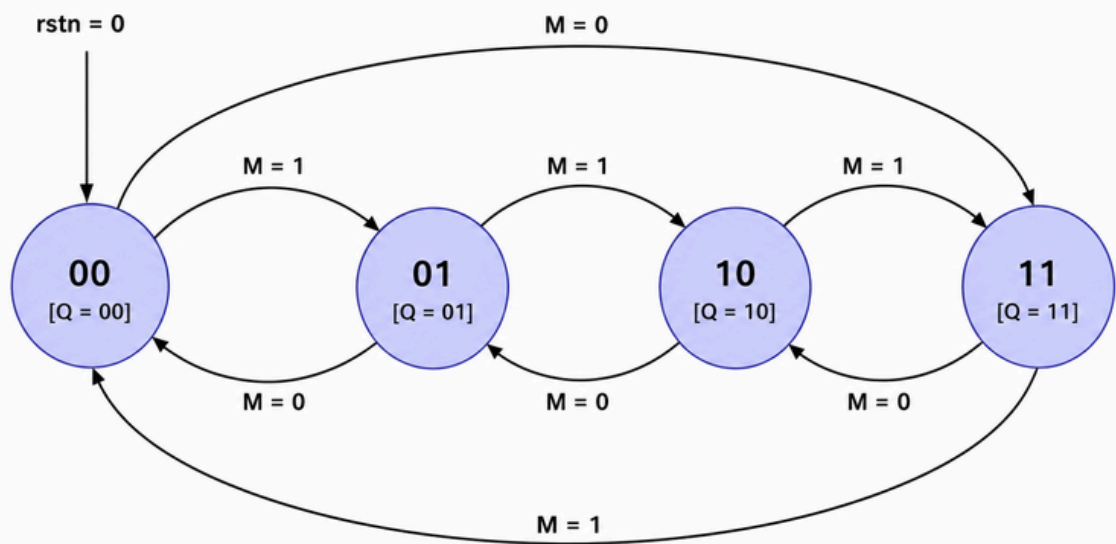
```
module fsm (  
    input logic clk,  
    input logic rstn,  
    input logic in,  
    output logic out  
);  
enum logic [1:0] {  
    A=2'b00,  
    B=2'b01,  
    C=2'b11,  
    unused = 2'bx  
} current_state, next_state;  
  
always_comb begin  
    case (current_state)  
        A: begin  
            if (in == 1'b0) begin  
                next_state = B;  
                out = 1'b0;  
            end  
        else begin  
            next_state = C;  
            out = 1'b0;  
        end  
    end  
  
    B: begin  
        if (in == 1'b0) begin  
            next_state = B;  
            out = 1'b1;  
        end  
    else begin  
        next_state = C;  
        out = 1'b0;  
    end  
end
```

```
        C: begin  
            if (in == 1'b0) begin  
                next_state = B;  
                out = 1'b0;  
            end  
        else begin  
            next_state = C;  
            out = 1'b1;  
        end  
    end  
  
    default: begin  
        next_state = unused;  
        out = unused;  
    end  
endcase  
end  
  
always @(posedge clk or negedge rstn)  
begin  
    if (!rstn)  
        current_state <= A;  
    else  
        current_state <= next_state;  
    end  
  
endmodule
```

Task #12.3

SystemVerilog Code:

```
module counter_2bit(  
    input logic clk,  
    input logic rstn,  
    input logic M,  
    output logic [1:0] Q  
);  
  
always @(posedge clk or negedge rstn)  
begin  
    if(!rstn)  
        Q <= 2'b00;  
  
    else if(M)  
        Q <= Q + 1;  
  
    else  
        Q <= Q - 1;  
end  
endmodule
```



M = 1 → Count Up

M = 0 → Count Down

Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task12.1	✓	✓	✓	✓	✓
Task12.2	✓	✓	✓	✓	✓
Task12.3	✓	✓	✓	✓	✓

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